Text Generation Using Google Gemini LLM API

Develop a Python program using the Gemini API to generate responses for different user prompts and display the generated output.

Aim

To develop a Python program that connects to the Google Gemini API, accepts different user prompts repeatedly, and displays the generated responses.

Algorithm

1. Import the Google Gen AI SDK and read the API key using hidden input.

2. Create a Gemini API client with the key.

3. Accept a prompt from the user inside a loop.

4. If the prompt is exit, end the program. If it is empty, request another prompt.

5. Send each valid prompt using client.models.generate_content().

6. Display the text returned in response.text.

7. Handle API or connection errors and continue accepting prompts.

Requirements and Colab Setup

Google Colab, internet access, and a Gemini API key with model access and available quota. Obtain the key from Google AI Studio:
https://aistudio.google.com/apikey

Run this command in the first Colab cell:
%pip install -q -U google-genai

Paste the Python program into a second cell and run it. Enter the API key in the hidden input field. Enter different prompts; type exit when finished.

Python Program

Each prompt is sent as an independent request. The loop accepts multiple prompts; it does not retain conversation history.

```
from getpass import getpass
from google import genai
from google.genai import errors

api_key = getpass("Enter Gemini API key: ").strip()
if not api_key:
    raise SystemExit("An API key is required.")

with genai.Client(api_key=api_key) as client:
    print("Type 'exit' to stop.")
    while True:
        prompt = input("\nEnter your prompt: ").strip()
        if prompt.lower() == "exit":
            print("Program ended.")
            break
        if not prompt:
            print("Please enter a non-empty prompt.")
            continue
        try:
            response = client.models.generate_content(
                model="gemini-3.8-flash",
                contents=prompt
            )
            print("Generated Output:")
            print(response.text or "No text returned.")
        except errors.APIError as error:
            print("API error:", error.code)
            print("Check key, model access and quota.")
        except Exception as error:
            print("Request failed:", type(error).__name__)
```

Input

Enter Gemini API key: [hidden input]
Prompt 1: Define AI in one sentence.
Prompt 2: List two uses of AI.
Final input: exit

Output: Colab-style Illustration

This image shows the expected display format and sample responses. It is a prepared illustration, not a screenshot of a live Colab execution. Actual responses may vary.

Colab-style output illustration

ILLUSTRATIVE - NOT A LIVE API RUN

```
[ ]  # Gemini text generation cell output

Enter Gemini API key: [hidden]
Type 'exit' to stop.

Enter your prompt: Define AI in one sentence.
Generated Output:
Artificial intelligence enables computers to perform tasks that
usually require human intelligence.

Enter your prompt: List two uses of AI.
Generated Output:
1. Detecting objects in images.
2. Recommending products to customers.

Enter your prompt: exit
Program ended.
```

Program Explanation

genai.Client(api_key=api_key) creates the client. The while loop accepts different prompts. generate_content() submits each prompt to the selected Gemini model, and response.text provides the text to display. The program handles empty input and ends when the user types exit.

Error Handling

API errors display a status code so the user can check authentication, model access, or quota. Other request failures display the exception type. If no text is returned, the program prints a message instead of displaying an empty response.

Result

A Python program was developed to generate and display responses for different user prompts using the Gemini API. The program was checked for valid Python syntax. Live API execution requires a valid key, available quota, and network access; the included output image is illustrative.

References

Official Google documentation:
1. Gemini API: Generating content
https://ai.google.dev/api/generate-content
2. Google Gen AI Python SDK
https://googleapis.github.io/python-genai/